

Excellent Performance of Few-Layer Borocarbonitrides as Anode Materials in Lithium-Ion Batteries

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Abstract: Borocarbonitrides ($B_xC_yN_z$) with a graphene-like structure exhibit a remarkable high lithium cyclability and current rate capability. The electrochemical performance of the $B_xC_yN_z$ materials, synthesized by using a simple solid-state synthesis route based on urea, was strongly dependent on the composition and surface area. Among the three compositions studied, the carbon-rich compound $B_{0.15}C_{0.73}N_{0.12}$ with the highest surface area showed an exceptional stability (over 100 cycles) and rate capability over widely varying current density values ($0.05\text{--}1\text{ A g}^{-1}$). $B_{0.15}C_{0.73}N_{0.12}$ has a very high specific capacity of 710 mA h g^{-1} at 0.05 A g^{-1} . With the inclusion of a suitable additive in the electrolyte, the specific capacity improved drastically, recording an impressive value of nearly 900 mA h g^{-1} at 0.05 A g^{-1} . It is believed that the solid-electrolyte interphase (SEI) layer at the interface of $B_xC_yN_z$ and electrolyte also plays a crucial role in the performance of the $B_xC_yN_z$.

High-performance rechargeable batteries, especially based on lithium intercalation chemistry, are now considered to be the most viable electric power storage device for large-scale applications.^[1,2] The most important criteria for large-scale applications of batteries are high energy and power density as well as safety. The first two criteria depend predominantly on the combination of cathode, anode, and electrolyte. Considerable work has focused on improving the performance of prevalent materials and developing materials with newer chemical compositions. On the anode front, due to the increased rate of electrolyte degradation with nanostructured carbons such as nanographite and fullerenes, several non-carbonaceous compounds (e.g., Ti-based and Sn-based compounds, and transition metal oxides) which exhibit promising lithium storage at widely varying current values

and enhanced safety have been explored.^[3–6] Most of the non-carbonaceous materials reversibly store lithium at voltages of typically $0.6\text{--}1.5\text{ V}$ ($\text{Li}^+|\text{Li}$), values much higher than that of graphite ($\approx 0.1\text{ V}$) and, therefore, the cell voltage of the full lithium-ion cell will decrease, which results in lower energy densities. Due to this drawback associated with non-carbonaceous materials, it is worthwhile to revisit the intercalation chemistry of carbonaceous compounds with newer chemical compositions that may display high electrochemical performance and cell safety.

There is a growing interest in the last few years in doping extended carbon structures like graphene with B and N in order to improve chemical, mechanical, and electrical properties.^[7] In this context, the interesting properties of borocarbonitrides ($B_xC_yN_z$) make them versatile for various applications.^[6,8,9] Few reports have demonstrated the use of heteroatom-doped graphene as anodes in lithium-ion batteries,^[10] and their cyclability is better compared to graphite, graphene, and other carbon nanostructures.^[6] Various synthesis routes^[11] such as gas-phase synthesis, solid-state synthesis, and vapor deposition (e.g., chemical vapor deposition (CVD)) have been employed to synthesize borocarbonitrides. Depending on the synthesis method, a wide range of morphologies such as layered structures, microspheres, and nanotubes have been obtained. A novel chemical method for the synthesis of $B_xC_yN_z$ with a high surface area is the solid-state route by Rao and co-workers.^[11a] $B_xC_yN_z$ synthesized by this solid-state route possess a layered structure consisting of 4–5 disordered graphene-like layers with an interlayer spacing of 0.44 nm . The disordered hexagonal layers of $B_xC_yN_z$ species are formed by B–C, B–N, C–N, and C–C bonds. The layers are thought to have BN islands dispersed in a graphene-like C–C network without any long range order between the layers along the c direction. The electrical properties of $B_xC_yN_z$ have been observed to depend on the number of insulating B–N bonds, various edge states, defects at the G–BN interface, and type of G–BN interfaces.

Figure 1 shows high-resolution transmission electron microscopy (HRTEM) images and the nitrogen adsorption-desorption isotherms of $B_{0.15}C_{0.73}N_{0.12}$. This compound has been chosen here as the representative among the three $B_xC_yN_z$ compositions ($B_{0.15}C_{0.73}N_{0.12}$, $B_{0.32}C_{0.42}N_{0.25}$, and $B_{0.43}C_{0.36}N_{0.26}$) for the above-mentioned characterization techniques. The specific surface area of this composition calculated using the BET method is observed to be the highest ($\approx 2000\text{ m}^2\text{ g}^{-1}$) compared to the two other compositions (both

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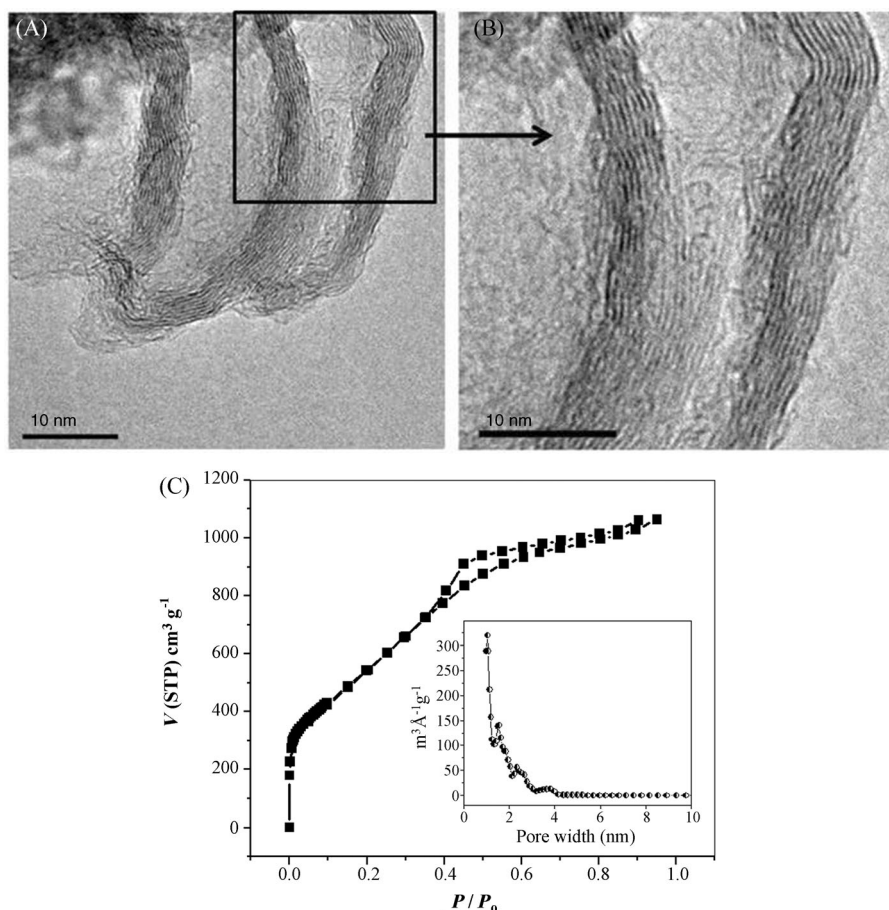


Figure 1. (A, B) HRTEM images of the B_{0.15}C_{0.73}N_{0.12} sample. (C) BET adsorption isotherm of B_{0.15}C_{0.73}N_{0.12}.

≈ 1300 m² g⁻¹). The isotherm shows a type IV behavior, thus implying the presence of mesopores inside the samples. The simultaneous presence of micropores and mesopores can also be observed from the adsorption isotherm and pore size distribution. The X-ray photoelectron spectroscopy (XPS) data (see the Supporting Information) reveal the presence of B–C, B–N, C–N, and C–C bonds (but no B–B or N–N bonds), which further confirm the existence of BN islands dispersed in a graphene like C–C network. The layered morphology can be further observed in the field-emission scanning electron microscopy (FESEM) images.

The cyclability of B_{0.15}C_{0.73}N_{0.12}, B_{0.32}C_{0.42}N_{0.25}, and B_{0.43}C_{0.36}N_{0.26} was evaluated at widely varying current values. Figure 2A shows the charge–discharge profiles at 0.05 A g⁻¹ for B_{0.15}C_{0.73}N_{0.12}. Similar charge–discharge profiles were also obtained for B_{0.43}C_{0.36}N_{0.26} and B_{0.32}C_{0.42}N_{0.25}. The profiles show very short plateaus in both charge and discharge processes. The discharge plateaus at around 0.9 V and 0.53 V are assigned to the formation of a solid–electrolyte interphase (SEI) and lithium intercalation into B_xC_yN_z layers, respectively.^[12,6] The intercalation phenomenon was also confirmed by a broad peak between 0.5 V and 1 V in cyclic voltammograms (Figure S6, Supporting Information) and derivative dq/dv versus v plots (Figure S7, Supporting Informa-

tion). The plateaus at around 1.2 V and 2.5 V observed in the charge cycle are due to the delithiation process from the B_xC_yN_z layers.^[10c,12] The disappearance of pronounced charge–discharge plateaus from 2nd cycle onwards can be accounted on the basis of the unique layered structure of B_xC_yN_z.^[10b,11] The intercalation and de-intercalation of Li ions are affected by the disordered layer stacking of B_xC_yN_z, edges, and topological defects at G–BN interfaces, which act as adsorption site for impurities and also lithium ions.^[11a,12] As a result, the prominent charge–discharge plateaus (Figure 2A) and evident peaks in the cyclic voltammograms disappear upon successive cycling. This may also be partially responsible for the capacity loss in the 1st charge cycle.

Plots of the specific capacity as a function of cycle number for B_{0.15}C_{0.73}N_{0.12}, B_{0.32}C_{0.42}N_{0.25}, and B_{0.43}C_{0.36}N_{0.26} cycled at a current density of 0.05 A g⁻¹ are shown in Figure 2B. Discharge capacities at the end of the first

cycle for B_{0.43}C_{0.36}N_{0.26}, B_{0.32}C_{0.42}N_{0.25}, and B_{0.15}C_{0.73}N_{0.12} samples are 1045 mA h g⁻¹, 1571 mA h g⁻¹ and 1855 mA h g⁻¹, with Coulombic efficiencies of 47 %, 43 %, and 40 %, respectively (Table S1, Supporting Information). The latter increase to almost 95 % from the 2nd cycle onwards, as shown in Figure 2D. A close look at the Coulombic efficiency shows that an increase in the B and N content in the graphene like carbon framework leads to an increase in Coulombic efficiency (Table S1, Supporting Information). This can be explained on the basis of a suppression of electrolytic degradation on the electrode surface due to the higher content of B and N in the graphene-like carbon framework.^[14] The low Coulombic efficiency in the initial cycle is due to the formation of an SEI that results in irreversible lithium loss.^[12] However, the SEI is stabilizes quickly and no further loss of lithium ions takes place. The charge–discharge capacities at the end of the 100th cycle for B_{0.43}C_{0.36}N_{0.26}, B_{0.32}C_{0.42}N_{0.25}, and B_{0.15}C_{0.73}N_{0.12} samples were 415 mA h g⁻¹, 470 mA h g⁻¹, and 675 mA h g⁻¹, respectively. The capacity value for B_{0.15}C_{0.73}N_{0.12} is remarkably high and is twice that of commercial graphite (≈ 372 mA h g⁻¹).^[10c] The specific capacity of B_{0.15}C_{0.73}N_{0.12} could be further improved by the addition of fluoroethylene carbonate (FEC) as an electrolyte additive to an ethylene carbonate (EC)/diethyl carbonate

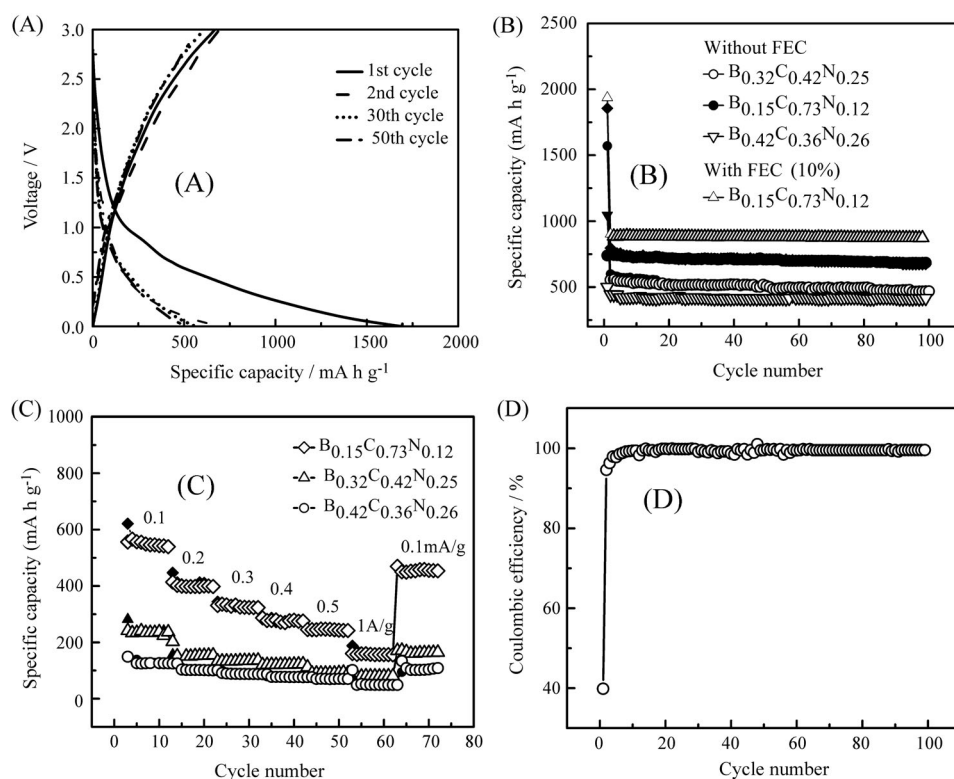


Figure 2. (A) Galvanostatic charge–discharge profiles for B_{0.15}C_{0.73}N_{0.12}. (B) Cycling at 0.05 A g^{−1} for the three B_xC_yN_z samples. (C) Current rate capabilities of B_xC_yN_z in the range of 0.1–1 A g^{−1}. (D) Coulombic efficiency of B_{0.15}C_{0.73}N_{0.12} as a function of cycle number.

(DEC, 1:1 v/v) solvent mixture. The specific capacity increased from 709 mA h g^{−1} (at 0% FEC) to nearly 900 mA h g^{−1} in the 40th cycle upon addition of 10 vol. % FEC. The enhanced capacity and improved cycling performance can be attributed to an even more stable SEI on the electrode/electrolyte surface and fast lithium transport through the electrode/electrolyte interface during successive cycling.^[15]

Figure 2C shows the current rate capabilities of the three B_xC_yN_z samples in the current density range of 0.1–1 A g^{−1}. It is observed that the cycling capacity of B_{0.15}C_{0.73}N_{0.12} is far superior compared to that of B_{0.43}C_{0.36}N_{0.26} and B_{0.32}C_{0.42}N_{0.25}. At the highest current density (1 A g^{−1}), a reversible capacity of 150 mA h g^{−1} is obtained. All samples recovered to their initial specific capacity values (at 0.1 A g^{−1}) following cycling at higher current values. The capacity of B_{0.15}C_{0.73}N_{0.12} at the relatively low current density of 0.3 A g^{−1} is higher than that of B_xC_yN_z nanosheets^[10c] that were shown to have the best battery performance among all known B_xC_yN_z compounds, whereas at the highest current rate (1 A g^{−1}), the capacity of B_{0.15}C_{0.73}N_{0.12} (150 mA h g^{−1}) is comparable to the previously reported values.^[10c] As all the samples possess the same morphology, the differences in battery performance can be correlated to the quantity of B and N in the graphene-like network and their surface areas. B_{0.15}C_{0.73}N_{0.12} showed the highest specific capacity as it possessed the highest surface area, thus implying a higher lithi-

um flux at the B_xC_yN_z/electrolyte interface that leads to higher lithium storage. As can be observed from Figure 2, the battery performance improves with an increase in carbon content in the order B_{0.43}C_{0.36}N_{0.26} < B_{0.32}C_{0.42}N_{0.25} < B_{0.15}C_{0.73}N_{0.12}. The results obtained from XPS analysis also provide an interesting correlation between the bonding of B, N, and C and the battery performance. For B_{0.15}C_{0.73}N_{0.12}, the percentage area for B–N bonding in B 1s (35%) is much less compared to that of B_{0.32}C_{0.42}N_{0.25} and B_{0.43}C_{0.36}N_{0.26} (65% and 73%, respectively; see the Supporting Information). The lower percentage area results in an increase in insulating BN domains (i.e., more B–N bonds than C–N and C–B bonds) and leads to a decrease in electronic conductivity of the graphene-like network, thus leading to a decrease in the capacity of B_{0.32}C_{0.42}N_{0.25} and B_{0.43}C_{0.36}N_{0.26}. The excellent electrochemical

performance of the reported materials has been supported by alternating current (AC) impedance spectroscopy (see the Supporting Information). Interestingly, although the capacity decreases with increasing B and N content, the Coulombic efficiency increases with increasing B and N content. This may be attributed to a suppression of the electrolytic degradation, as discussed previously.^[14]

We have successfully demonstrated here that few-layer borocarbonitrides B_xC_yN_z, synthesized by the solid-state route based on urea, can be effectively employed in lithium batteries as a promising alternative to graphite anodes. The cycling efficiency at varying currents is excellent and is observed to be strongly dependent on the chemical composition and surface area of B_xC_yN_z. Materials with higher carbon content and a higher surface area exhibit excellent cyclability and rate capability, and are even superior to the BCN-type materials reported in the literature at certain current rates. It is envisaged that additional improvements (i.e., the use of other electrolyte additives and composite electrode composition) may further enhance the performance of borocarbonitrides.

Experimental Section

Borocarbonitrides (B_xC_yN_z) were prepared by the urea-based solid-state synthesis previously reported.^[11a,13] In a typical synthesis, boric acid, high-surface-area activated charcoal, and urea were mixed at definite propor-

tions in water and heated at 80°C. The slurry was then transferred to a quartz boat and heated first at 900°C for 10 h under a nitrogen atmosphere followed by heating under an NH₃ atmosphere at the same temperature. In the present work, three different compositions of B_xC_yN_z, i.e., B_{0.15}C_{0.73}N_{0.12}, B_{0.32}C_{0.42}N_{0.25}, and B_{0.43}C_{0.31}N_{0.26} were synthesized. The compositions were estimated using X-ray photoelectron spectroscopy (XPS). The as-prepared samples were characterized by transmission electron spectroscopy (TEM, Tecnai T20 instrument (FEI)) operated at an accelerating voltage of 200 kV. Specific surface areas were measured by Brunauer–Emmett–Teller (BET) nitrogen adsorption–desorption (Quanta Chrome Autosorb-1). For measurements of N₂ adsorption–desorption isotherms, samples were degassed at 140°C for 4 h prior to the measurements. Pore size distributions were calculated from the desorption branch of the N₂ adsorption isotherm using the Barrett–Joyner–Halenda (BJH) formula at 77 K. For finding the binding energies associated with various chemical states and composition, X-ray photoelectron spectroscopy (XPS; Thermo-Scientific Multilab 2000; Al_{Kα} X-rays) data were recorded. Binding energies are reported with reference to C 1s at 284.5 eV and are accurate within ±0.1 eV. Electrochemical characterization of the layered B_xC_yN_z compositions was performed in home-built Swagelok cells, in which pure metallic Li (Aldrich) acted as both counter and reference electrodes and B_xC_yN_z was taken as the working electrode. The electrodes were prepared by mixing the anode material BCN with carbon black and PVDF binder (80:1:1 w/w/w) in *N*-methyl-pyrrolidone. The formed slurry was spread uniformly on the copper current collector and dried at 120°C in vacuum overnight. The loaded mass of the active material was 1.6 mg. A solution of LiPF₆ (1 M) in EC/DEC (1:1 v/v) was used as the electrolyte. Galvanostatic charge–discharge cycling (Arbin Instruments, MSTAT) was carried out at room temperature at various current densities (0.05–1 A g^{−1}) within the voltage range of 0–3 V.

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